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Coupling intermittent micro-aeration to anaerobic digestion improves tetra-azo dye Direct Black 22 treatment in sequencing batch reactors

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ABSTRACT

The effects of discontinuous micro-aeration on the biotransformation of the tetra-azo dye Direct Black 22 (DB22) and of its anaerobic by-products (aromatic amines) were tested using sequencing batch reactors. The experimental set-up consisted of three independent reactors (R1, R2, and R3) working with distinct cycles: R1 was completely anaerobic, while both R2 and R3 included a micro-aerated stage after an anaerobic stage (R2 continuously and R3 intermittently micro-aerated). Average DB22 removal efficiencies in R1, R2, and R3 were 81.4, 74.5 and 76.8%, respectively. R1 produced an effluent with high acute ecotoxicity (1.13% as effective concentration causing 50% of bioluminescence inhibition in *Vibrio fischeri*) due to aromatic amines accumulation. In contrast, no acute ecotoxicity was detected in the effluent of R2 and R3, reflecting aromatic amines removal. At the anaerobic stage of the cycle, R2 and R3 showed DB22 decolorization rates 73 and 123% higher than R1, respectively. At the same time, the corresponding values of estimated methane production were 35 and 37% lower. Thus, oxygen apparently changed the balance between methanogenesis and reductive decolorization, inhibiting the first process and enhancing the second.

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1. Introduction

In textile industry, dye fixation to fabrics occurs only partially (50–98%), generating wastewater with high coloration. The main pigments used in this sector are azo dyes (about 70%), which are featured by the presence of azo bonds (—N═N—). The azo dyes and their anaerobic degradation by-products can be mutagenic, carcinogenic, and toxic to the environment. High coloration may also inhibit photosynthesis in water bodies (van der Zee and Villaverde, 2005; dos Santos et al., 2007).

To solve this problem, biological treatment stands out as a low-cost environmental-friendly alternative that can be used in industries of diverse scales. Generally, biological treatment removes azo dyes and their by-products from textile wastewater by a combination of anaerobic and aerobic processes (Field et al., 1995). An initial anaerobic step is required since azo dyes are known to generally remain unaffected under aerobic conditions (Pagga and Brown, 1986; Shaul et al., 1991). The azo bonds are highly electrophilic, which protects the dye molecule against attacks by oxygenase enzymes (Kulla et al., 1983; Field et al., 1995).

On the other hand, the anaerobic cleavage of azo bonds, the chromophore groups, reduces the effluent coloration through the action of non-specific exoenzymes (azoreductases). The dye acts as an electron acceptor during the degradation of an organic substrate. Aromatic amines are formed after the reduction of azo bonds. Such compounds are colorless, although they can be highly toxic. These amines are then degraded aerobically, acting as energy and carbon source (van der Zee and Villaverde, 2005; Khan et al., 2013; Arora, 2015).

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Due to what has been mentioned, the use of two treatment units (anaerobic followed by aerobic) is frequent (Jonstrup et al., 2011; da Silva et al., 2012; Amaral et al., 2014; Baêta et al., 2015). Nevertheless, anaerobic and aerobic mixed microbial cultures may be able to reduce azo bonds and degrade aromatic amines when in the absence and presence of oxygen, respectively (Field et al., 1995). This suggests the development of a compact treatment in a single unit subject to the two conditions (anaerobic and aerobic) over time. It was also observed that micro-aeration, under 5–10% of the oxygen saturation concentration, may be enough to mineralize most of the biodegradable aromatic compounds (Field et al., 1995; Gavazza et al., 2015; Kumar et al., 2015).

In a practical approach, sequencing batch reactors (SBR) are an appropriate alternative for the treatment of intermittently produced wastewater. This is due to the hydraulic regime of these reactors, which do not work with a continuous feed, by their very definition. Additionally, reactors without continuous discharge may accumulate oxygen in the aerated phases, causing unnecessary electric energy consumption in treatment units. Intermittent aeration can be an alternative to keep low concentrations of oxygen in this case. However, few studies have been published regarding the combination of anaerobic and micro-aerobic conditions for azo dye degradation in the same compartment (Gavazza et al., 2015; Balapure et al., 2016). Even fewer tested the use of intermittent micro-aeration (Kumar et al., 2015).

With this background, this study aimed to evaluate the effects of continuous and intermittent micro-aeration, both applied after an anaerobic step, on tetra-azo dye Direct Black 22 (DB22) and its by-products transformation, using biological sequencing batch reactors.

2. Materials and methods

2.1. Synthetic wastewater

Tetra-azo dye Direct Black 22 (DB22; $C_{44}H_{32}N_{13}Na_3O_{11}S_3$; molecular weight: 1083.97 g mol^{-1}; CAS # 6473-13-8) was obtained from ExataCor©. Dye purity was not provided, thus nuclear magnetic resonance (NMR) spectroscopy was applied for its determination before starting the experiment, as described in the Supplementary material. DB22 purity was found to be 50% on a mass basis.

Synthetic effluent was designed to reproduce real textile wastewater, following recommendation of Amorim et al. (2013). It was composed of: DB22 (0.03 mM), ethanol (1200 mg L^{-1} of Chemical Oxygen Demand — COD), $NaHCO_3$ as buffer (1200 mg L^{-1}), NaCl as a usual interfering substance in textile effluents (1000 mg L^{-1}), macro and micronutrients. DB22 dissolution in water followed the procedures of Gavazza et al. (2015). Ethanol is not a common component of real textile wastewaters. Starch is more common. However, for practical reasons, ethanol was chosen as electron donor for the azo dye reduction in the present work.

The macronutrients concentrations in the synthetic effluent were (mg L^{-1}): NH_4Cl (280), K_2HPO_4 (252), $MgSO_4{\cdot}7H_2O$ (100) and $CaCl_2$ (7), adapted from dos Santos et al. (2006d). The amount of 1 mL L^{-1} of micronutrients solution (Florêncio et al., 1993) was added to the synthetic effluent and it was composed of (mg L^{-1}): $FeCl_2{\cdot}4H_2O$ (2000), H_3BO_3 (50), $ZnCl_2$ (50), $MnCl_2{\cdot}4H_2O$ (500), $CuCl_2{\cdot}2H_2O$ (38), $(NH_4)Mo_7O_{24}{\cdot}4H_2O$ (50), $AlCl_3{\cdot}6H_2O$ (90), $CoCl_2{\cdot}6H_2O$ (2000), $NiCl_2{\cdot}6H_2O$ (142), $NaSeO_3{\cdot}5H_2O$ (164), EDTA (1000) and HCl (1 mL L^{-1}). Resazurin was not added because of its properties as a redox mediator (dos Santos et al., 2006).

2.2. Experimental set-up

The experimental set-up consisted of three independent sequencing batch reactors (R1, R2, and R3) working with distinct anaerobic/micro-aerobic cycle operating conditions. Cylindrical acrylic reactors (20 cm of internal diameter and 19 cm of height) with 5 L of working volume (replacement volume of 50%) were operated at mesophilic temperature of 37 ± 1 °C for 107 days.

In each reactor, the cycle time was 24 h distributed between: filling (15 min), reacting (23 h), settling (30 min), and drawling (15 min). The 23 h of reaction time in R1 was completely anaerobic. R2 combined 12 h of anaerobic stage and 11 h of micro-aerated stage in the reaction time. R3 operated similarly to R2, with intermittent air supply (30 min every 2 h) in the 11 h of the micro-aerated stage.

Circular aerators made of micro-perforated rubber (Resun, Air Curtain, Raoping, China), placed at the bottom of R2 and R3, were responsible for sparging the air at a flow rate of 0.25 ± 0.05 L min^{-1} (measured with a Tecnofluid flowmeter, TRV-140-K-7-P, Nova Lima, Brazil) producing few fine bubbles. Timers (Clip, CLB-40, São Roque, Brazil) were used to control the power supply for the aerator (Jad, Air-Pump S-2000, Raoping, China) in R2 and R3.

Anaerobic sludge from a pilot scale UASB reactor treating real textile wastewater, previously operated by Amaral et al. (2014), was used as inoculum (2.5 gVSS L^{-1} in the mixed liquor). Substrate-biomass contact was ensured by magnetic stirrers (Quimis, Q-241, Diadema, Brazil), during the reaction phase.

2.3. Analytical procedure

Samples taken from the outlet of the reactors at the end of the cycle time were analyzed three times a week by measuring pH, alkalinity, oxidation-reduction potential (ORP), dissolved oxygen (DO), salinity, COD, DB22 concentration, and aromatic amines (qualitatively). Volatile suspended solids (VSS) in the effluent were monitored fortnightly. VSS were also measured in the mixed liquor on the first and on the last operational days.

Temporal concentration profiles were performed after the reactors reached apparent steady-state conditions. The measured parameters were: ORP, DO, ethanol, volatile fatty acids (VFA), DB22 concentration, and aromatic amines (qualitatively) over the cycle time of 24 h.

DB22 and aromatic amines were monitored spectrophotometrically (Hitachi U-2910, Hitachi, Ltd., Tokyo, Japan). Samples were centrifuged for 15 min at 3500 rpm and the clarified supernatants were diluted to bring their light absorbance to values lower than 0.9 in the wavelengths of interest. DB22 maximum wavelength of light absorption is 476 nm, which was used for measuring its concentration. The molar absorption coefficient, i.e. a factor to convert absorbance measurements into concentration, was 56.2 cm^{-1} mM^{-1}.

Qualitative analysis of aromatic amines was performed by scanning the samples in the light absorption range of these compounds (200–350 nm) (Pinheiro et al., 2004). This method is commonly used for that purpose (e.g. Jonstrup et al., 2011; Amaral et al., 2014; Baêta et al., 2015).

Ethanol and VFA were analyzed by a gas chromatograph-GC (Agilent Technologies 7890 A, Agilent Technologies Inc., Wilmington, DE, USA). The GC column J&W DB-WAXERT (30 m × 250 μm × 0.25 μm) was used for ethanol analysis under the following operating conditions: inlet volume of 1.0 μL; splitless for ethanol concentration from 10 to 100 mg L^{-1}, and split of 100:1 for ethanol concentration from 200 to 1000 mg L^{-1}; hydrogen flow rate of 1 mL min^{-1}; FID detector

at 300 °C; inlet at 250 °C; and oven ramp temperature program with initial temperature of 30 °C for 2 min, temperature ramp of + 2 °C min^{-1} until 40 °C for 1 min. The same column was used for VFA analysis, under the corresponding conditions: inlet volume of 2.0 μL; split 1:10 as inlet method; hydrogen flow rate of 1 mL min^{-1}; FID detector at 300 °C; inlet at 250 °C; and oven ramp temperature program with initial temperature of 100 °C for 1 min, temperature ramp of +8 °C min^{-1} until 150 °C for 1 min and +35 °C min^{-1} until 200 °C for 1 min.

Acute ecotoxicity of the influent and effluent of each reactor was evaluated at the end of the operational period by Microtox® (Microtox Model 500 analyzer, Strategic Diagnostics, Inc. SDIX, Newark, DE, USA), applying the bioluminescent bacterium *Vibrio fischeri* as the toxicity indicator (incubation time of 15 min). The measured parameters were the effective nominal concentrations (EC) resulting in 20 and 50% inhibition of bioluminescence and the minimum dilution causing no quantifiable inhibition.

The other parameters (pH, alkalinity, ORP, DO, salinity, COD, and VSS) were determined according to Standard Methods (APHA, 2005), using a 50-ml sample collected from the effluent without dilution.

2.4. Data analysis

Student's t test with 95% confidence intervals was applied to verify statistical difference between the reactors' performances throughout the experimental period.

Cell yield coefficient Y (kgVSS kgCOD^{-1}), the ratio between bacterial growth and total COD consumption in each reactor, was calculated accordingly Eq. (1). The sludge age of each reactor was estimated by Eq. (2).

$$Y = \frac{(VSS_{final\ ML} - VSS_{initial\ ML}).V_r - VSS_{efl}.V_{efl}.T}{COD_{consumed}.T} \quad (1)$$

$$\text{Sludge age} = \frac{(VSS_{initial\ ML} + VSS_{final\ ML}).V_r}{2.VSS_{efl}.Q_{efl}} \quad (2)$$

where $VSS_{initialML}$ (mg L^{-1}) is the initial concentration of VSS in the mixed liquor; $VSS_{finalML}$ (mg L^{-1}) is the final concentration of VSS in the mixed liquor; V_r (L) is the working volume of the reactor; VSS_{efl} (mg L^{-1}) is the average concentration of VSS in the effluent; T (d) is the duration of the operational period; COD_{cons} (mg) is the consumed COD average during 1 day; Q_{efl} (L d^{-1}) is the daily volume of effluent produced by the reactor.

Principal component analysis — PCA (Bro and Smilde, 2014) was applied to the spectral data (200–350 nm) of the temporal profiles as an exploratory tool to assess the formation and consumption of aromatic amines. Data pre-treatment and a PCA model were performed using The Unscrambler® (CAMO S.A.) software version 9.7. Savitzky–Golay smoothing (zero order polynomial and 7 points window) plus baseline offset were used as pre-treatments to reduce interferences and keep the relevant information in the spectral set. Data were mean centered, and full cross validation was applied during the modeling. Auto-scaling, routinely applied in discrete data sets before PCA analysis, is not recommended for spectral data to avoid masking stronger signals in important spectral regions (Beebe et al., 1998). First principal component (PC1) explained 73% of the total data variance and it was selected as the concentration indicator of aromatic amines.

Data from the cycle temporal profiles were used to calculate the rate constants for COD and DB22 consumption. Details about this calculation can be found in the Supplementary Material. COD data from the temporal profile were obtained by converting ethanol and VFA concentrations, as described by Eq. (3).

$$COD_{ethanol+VFA} = 2.08\,[ethanol] + 1.07\,[acetic\ acid] + 1.51[propionic\ acid] \quad (3)$$

where $COD_{ethanol+VFA}$ (mg L^{-1}) is the COD from ethanol and VFA; [*ethanol*], [*acetic acid*], and [*propionic acid*] (mg L^{-1}) are the concentrations of these compounds. The results from Eq. (3) were applied to determine $(k_0)_{COD}$ (mg L^{-1} h^{-1}), which is the zero-order kinetic constant for COD consumption (from ethanol and VFA).

Eq. (4) estimates the COD utilized for DB22 reductive decolorization along the cycle time (t) using a first-order kinetic model. Because DB22 is a tetra-azo dye, the reductive decolorization of 1 mmol of DB22 requires, theoretically, 16 mmol of electrons (4 for each azo bond), which represents 128 mg of COD (8 mg of COD per mmol of electrons) (Amaral et al., 2017). This is also accounted in Eq. (4).

$$COD_{decolorization} = 128\,[DB22]_{initial}(1 - e^{(k_1)_{DB22}.t}) \quad (4)$$

where $COD_{decolorization}$ (mg L^{-1}) is the COD consumed by DB22 decolorization during the cycle time; $[DB22]_{initial}$ (mM) is to the initial concentration of DB22 in a cycle; $(k_1)_{DB22}$ is the first-order rate constant (h^{-1}) for DB22 consumption; t (h) is the cycle time.

The maximum consumption of COD (5.76 mg L^{-1}) that would come from sulfate reduction (COD/sulfate ratio of 0.67) (Muyzer and Stams, 2008) of aerobically oxidized sulfur from DB22 molecules was neglected in this study. However, due to the use of magnesium sulfate in the macronutrients solution, sulfate reduction could have account for up to 7.7% of the COD consumption during the anaerobic stage of the cycles; therefore, it was considered in further calculations.

3. Results and discussion

3.1. Dye and COD removal

Apparent steady state conditions for dye and COD removal were established after 58 days of reactor operation. Within these conditions, the average of COD removal (confidence interval from Student's t test) in R1, R2, and R3 was 76.4 (75.1–77.7), 79.5 (76.7–82.2), and 81.4% (78.8–84.0), respectively. The corresponding values for DB22 removal were 81.4 (77.6–85.2), 74.5 (68.5–80.6) and 76.8% (73.4–80.1). The reactors behaved similarly for both COD (Fig. 1A) and dye (Fig. 1B) removal, although the 95% confidence intervals of the Student's t test indicated that R3 performed slightly better than R1 for COD removal only.

During the steady-state conditions, the micro-aerated reactors (R2 and R3) were able to degrade all the carbon source (ethanol and VFAs formed), causing a higher COD removal efficiency than that observed for R1, which accumulated acetic acid between cycles, as discussed later.

No lag phase was observed for dye degradation, presumably due to the inoculum source. However, lower variability in dye and COD concentrations was observed during steady state (Fig. 1B).

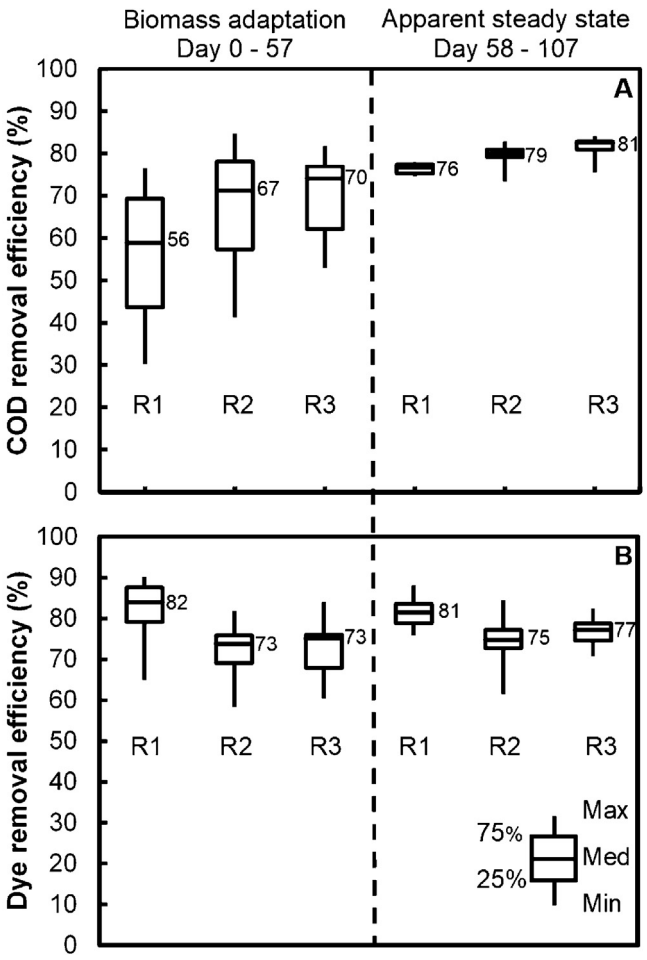


Fig. 1 – COD (panel A) and azo dye (panel B) removal efficiencies during biomass adaptation period (days 0–57, with 15 data points) and apparent steady state period (days 58–107, with 7 data points).

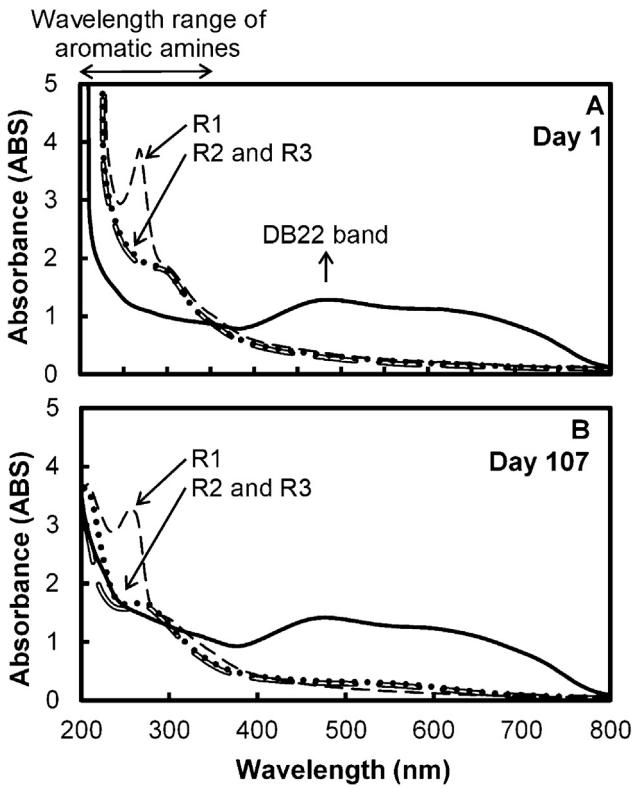


Fig. 2 – UV–vis spectrophotometer scans of influent (——) and effluents of R1 (— —), R2 (·····) and R3 (═) on days 1 (panel A) and 107 (panel B). Absorbances values are multiplied by the dilution factor of the sample.

The use of the sequencing batches herein allowed achieving similar removal efficiencies but using a much shorter reaction time when compared to those of Amorim et al. (2013). These authors used ethanol as an electron donor in single anaerobic batches of 120–264 h and observed 83.3% of DB22 removal even when redox mediators were applied, while in the present work we reached 81% (R1) with a cycle time of only 24 h. Furthermore, using continuous UASB reactors, Gavazza et al. (2015) observed DB22 removal efficiencies below 70% (hydraulic retention time — HRT of 16 h).

The performance of reactors R2 and R3 was similar to the results from other studies using single chamber anaerobic–aerobic reactors. Lourenço et al. (2001) obtained 75% removal efficiency of a diazo (Remazol Black B) and 90% for a monoazo dye (Remazol Brilliant Violet 5R) in anaerobic-aerobic SBRs with 24 h of HRT. Better results were described by Farabegoli et al. (2010), who observed color removal of 97% for a monoazo dye (Reactive Red 195).

3.2. *Aromatic amines*

The main DB22 absorbance band occurs in the visible spectrum around 476 nm, while the colorless aromatic amines absorb UV light from 200 to 350 nm (Pinheiro et al., 2004). Fig. 2 shows UV–vis spectrophotometer scan (200–800 nm) of samples in the first (Fig. 2A) and in the last day of operation (Fig. 2B). The absorbance band of DB22 (around 476 nm) strongly decreased in the effluent of all reactors in comparison with that of the influent. Hence, aromatic amines were formed in R1, R2, and R3, as seen by the absorbance increase in the 200–350 nm range. Since the first day and throughout the operating period, R1 produced and accumulated higher concentrations of aromatic amines than the other reactors. Apparently, R2 and R3 were able to remove part of the amines also since the first batch, what presumably happened under oxygen exposure. From the 72nd day of operation on the absorbance values in the 200–350 nm range for the effluents of R2 and R3 became close to the values observed for the influent, thus indicating consistent removal of aromatic amines in R2 and R3.

Benzene and naphthalene-based aromatic amines, with or without sulfonic groups, are putative products after the cleavage of azo bonds (Fig. 3). Some aromatic amines are known to be susceptible to aerobic degradation (Field et al., 1995; Fuchs et al., 2011; Arora, 2015).

3.3. *Ecotoxicity*

No acute ecotoxicity was identified in the influent (containing DB22, without aromatic amines) and in the effluents of the reactors supplied with oxygen (R2 and R3). However, strong acute toxicity was detected in the R1 effluent possibly due to the accumulation of aromatic amines. For R1, 0.41% ($EC_{20\%}$) of the effluent diluted in distilled water was enough to cause 20% of bioluminescence inhibition, while 1.13% ($EC_{50\%}$) inhibited 50% of the bioluminescence. The minimum concentration with no quantifiable toxicity was 0.39%. Then, these results confirm the need of combining anaerobic and, at least, microaerobic processes for DB22 degradation. Another important

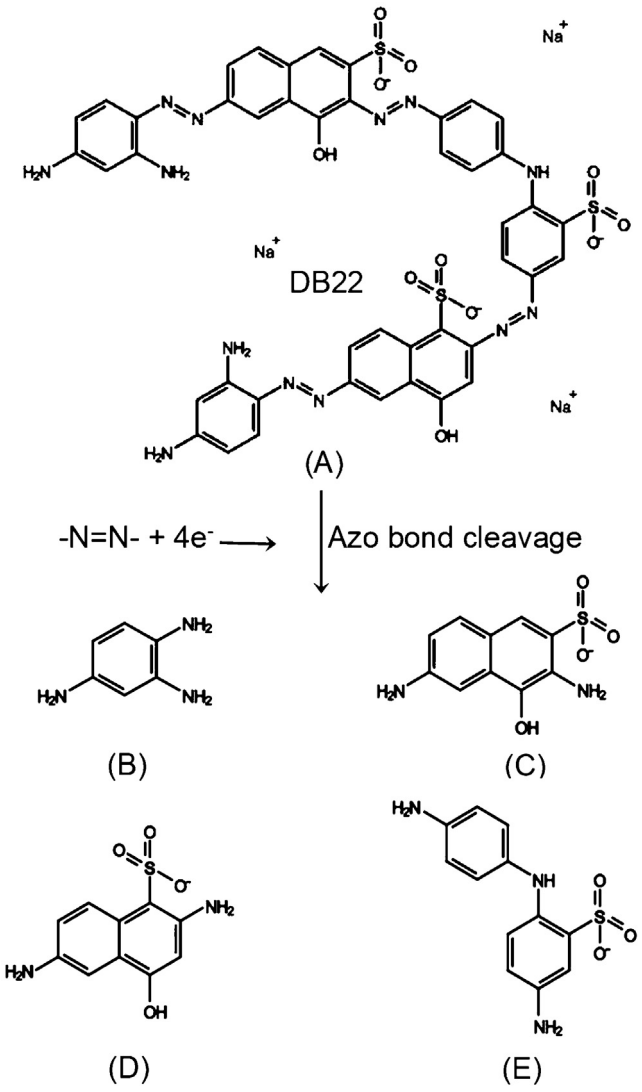


Fig. 3 – Structures of DB22 (A) and candidate metabolites from its anaerobic reduction: 1,2,4-Benzenetriamine (B), 3,6-diamino-4-hidroxy-naphtalene-2-sulfonilic acid (C), 2,6-diamino-4-hidroxy-naphtalene-1-sulfonilic acid (D), and 5-amino-2-(4-aminoanilino)benzene-1-sulfonic acid (E).

finding is that intermittent micro-aeration promoted complete removal of acute toxicity, as well as the continuous micro-aeration, but using only 25% of the oxygen.

Our results are consistent with those of Frijters et al. (2006), using Vibrio fischeri, and of Amaral et al. (2014), using Daphnia magna, both treating real textile wastewaters by the combination of anaerobic and conventional aerobic process. Those authors reported toxicity increase during the anaerobic step, followed by toxicity decrease after the aerobic step. Diverging from our observations, Franciscon et al. (2009a), using a facultative *Klebsiella* sp. treating several azo dyes, did not achieve the removal of aromatic amines and toxicity (Daphnia magna) after the micro-aeration process. Instead, they reported a decrease in toxicity only when conventional aerobic conditions were supplied. However, when using another culture, a facultative *Staphylococcus arlettae*, acute toxicity was lower after micro-aeration, but complete removal was only detected also under conventional aerobic conditions (Franciscon et al., 2009b).

The lack of toxicity found for DB22 in the influent deserves attention. A compound toxicity is usually proportional to its permeability into the cells, which is a function of molecular weight and lipid solubility (Rawat et al., 2016). Sulfonated groups in the structure (Fig. 3A) and the high molecular weight of DB22 were probably the factors responsible for the low permeability of this dye into *V. fischeri* cells, thus explaining the lack of toxicity for DB22.

The putative aromatic amines products from anaerobic degradation of DB22 include a non-sulfonated benzene-based aromatic amine, benzene-1,2,4-triamine (Fig. 3B), which is normally toxic (Rawat et al., 2016). The other amines, 3,6-diamino-4-hydroxynaphtalene-2-sulfonic acid (Fig. 3C), 2,6-diamino-4-hydroxynaphtalene-1-sulfonic acid (Fig. 3D), and 5-amino-2-(4-aminoanilino)benzene-1-sulfonic acid (Fig. 3E), typically exhibit lower toxicity than their analogs without sulfonic groups (Gottlieb et al., 2003). In addition, despite the weak effect of DB22 and of some aromatic amines to *Vibrio fischeri* cells, prolonged exposure to these compounds may cause severe DNA damage (Ferraz et al., 2011).

3.4. Cell yield coefficient

The cell yield coefficient (Y) was totally consistent with the operational conditions, with values of 0.0301, 0.0476, and 0.0441 kgVSS kgCOD^{-1} for R1, R2, and R3, respectively. Anaerobic degradation of organic matter via methanogenesis has a substantially lower energy yield than aerobic respiration. This explains the greater Y found for R2 and R3.

The cellular production in the anaerobic reactor (R1) was higher than that observed by Amaral et al. (2014) (0.0125 kgSSV kgCOD^{-1}) in a pilot UASB reactor used for treatment of real textile effluent. The authors reported that the release of many chemicals in real textile effluent slowed down the performance of the biological process, leading to a low Y value.

The sludge age was not controlled in the present work; the obtained values for R1, R2, and R3 of 328.1, 164.7, and 169.4 days, respectively, reflect the higher loss of sludge in the effluent of the micro-aerated reactors due to the higher cellular production.

3.5. Concentration profiles over one cycle

3.5.1. ORP and DO

During the first 12 h of cycle time in R1, R2, and R3, the ORP ranged between −535 and −450 mV (Fig. 4A), which is low enough to allow the co-existence of reductive decolorization of DB22 and methanogenesis. In R1, this range of ORP lasted the entire cycle time. For reduction of azo bonds, the ORP of the reducing equivalent (e.g. azoreductase or redox mediator) must be lower than the ORP of the azo dye itself, making electron transfer for azo bond cleavage possible. In anaerobic-aerobic systems, Lourenço et al. (2001) observed that the occurrence of ORP lower than −400 mV was required for effective reduction of azo bonds. Other authors suggest a less restrictive range (−100 to −500 mV) as the usual one for reductive decolorization of azo dyes (dos Santos et al., 2007d). At the same time, in methanogenic reactors, the ORP is referred to be lower than −300 mV (Harper and Pohland, 1986).

In R2, the continuous micro-aeration, which started from 12 h of cycle time, raised the ORP slowly from −535 (12.0 h) to 0 (19.5 h), reaching a maximum of +23 mV (20.5 h). The ORP behavior in R2 is linked to the organic matter consumption

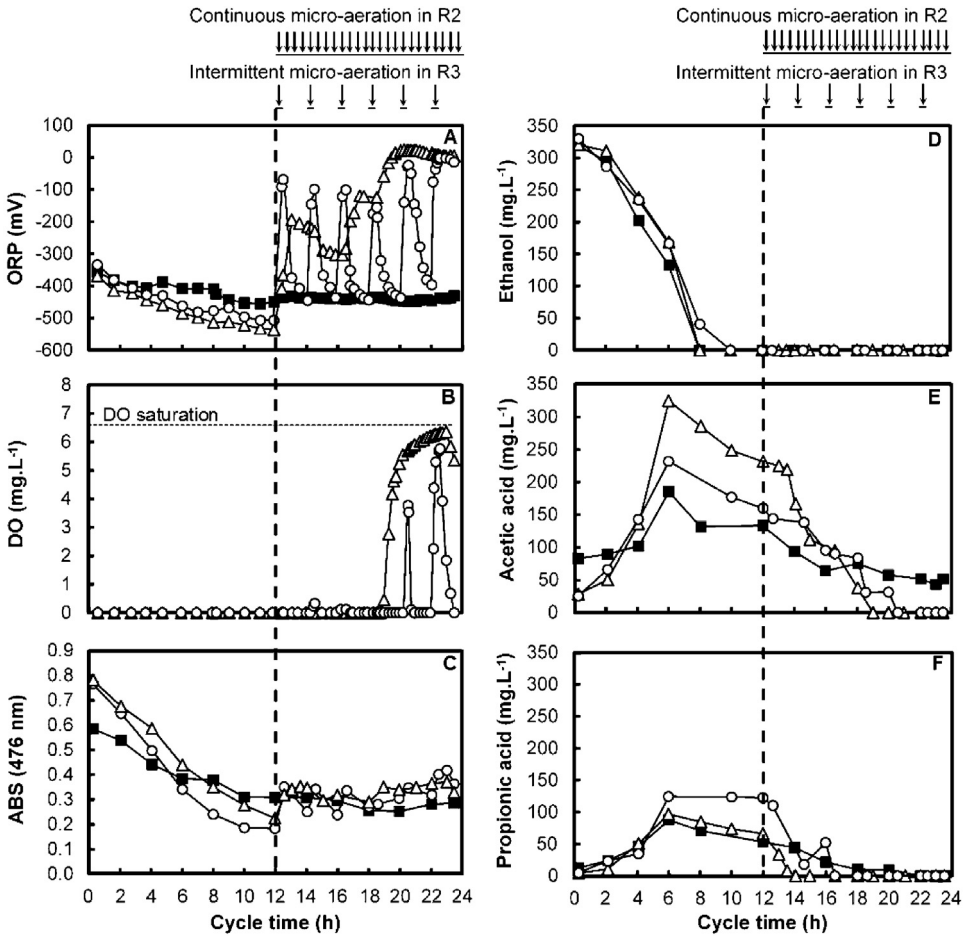


Fig. 4 – ORP (panel A), DO (panel B), absorbance at 476 nm (indicative of DB22) (panel C), ethanol (panel D), acetic acid (panel E), and propionic acid (panel F) of a typical steady-state cycle (day 98) in R1 (—■—), R2 (—△—) and R3 (—○—).

and reflects the oxygen accumulation at the end of the batch cycle, when the organic matter became limiting (from 19 h).

In R3, ORP ranged from −445 (14.0 h) to −2 mV (23.0 h), but steeply rose during micro-aeration periods, as well as steeply declined during the intervals without micro-aeration. The immediate consumption of the supplied DO (Fig. 4B) by the microbial community was responsible for the reduction of the ORP, as previously reported (Kato et al., 1997).

DO concentrations were below detection during the first few hours of micro-aeration in R2 (12.0–19.0 h) and R3 (12.0–20.3 h), indicating immediate DO consumption by the microbial community. Then, DO started to increase shortly after the total consumption of biodegradable organic matter (ethanol, VFA and presumably some aromatic amines) in R2 and R3 (Fig. 4D–F). DO concentrations became close to saturation (calculated for 37 °C and considering the salinity of the effluents) at 23.0 and 22.5 h for R2 and R3, respectively (Fig. 4B).

3.5.2. *Ethanol and VFA concentrations*

Ethanol (Fig. 4D) was completely consumed in the anaerobic phases, in 8.0 h for R2 and R1 and in 10 h for R3. Then, we searched for the following VFA: acetic, propanoic, isobutyric, butyric, isovaleric, valeric and caproic acids. Two VFA were identified in all reactors: acetic and propionic acids (Fig. 4E and F). Acetic acid (52 mg L^{-1}) accumulated in the final effluent of R1, while being rapidly and completely consumed in R2 and R3, from the beginning of the micro-aeration stages.

Methanogenesis, reductive decolorization, and aerobic respiration were presumably the main electron sinks for ethanol and VFA oxidation. In terms of COD, an assumed complete azo bond reduction would account for only 0.7, 0.9, and 1.0% of the total COD consumption in the complete cycle of R1 and in the anaerobic phases of R2 and R3, respectively. Methanogenesis would be correspondingly responsible for 91.6 to 99.3, 42.1 to 49.8 and 41.9 to 49.6% of COD consumption in R1, R2 and R3, respectively, considering that the consumption caused by sulfate reduction could have accounted for up to 7.7%. The aerobic respiration quota was 49.3 and 49.4% in R2 and R3, respectively. Higher contributions of anaerobiosis to COD removal from synthetic textile effluent in anaerobic-aerobic systems were reported by Sponza and Işik (2005) and Yaşar et al. (2012). Conversely, Ong et al. (2005) obtained greater contribution of aerobic degradation. Amaral et al. (2014) observed the same COD removal efficiencies in the anaerobic and aerobic units used for the treatment of real textile wastewater. In the present study, with the same microbial community exposed to anaerobic and micro-aerobic conditions in R2 and R3, methanogenesis and aerobic degradation were equally important for COD removal, with no differences between continuous and intermittent micro-aeration processes.

3.5.3. *DB22 and aromatic amines*

Fig. 3 shows the formation of aromatic amines from the reduction of DB22. Fig. 5 shows the probable formation and degradation of the aromatic amines through the lens of a PCA, which summed up the absorbance in the 200–350 nm range for each sample. First principal component (PC1) represented better the behavior of the absorbance in the range

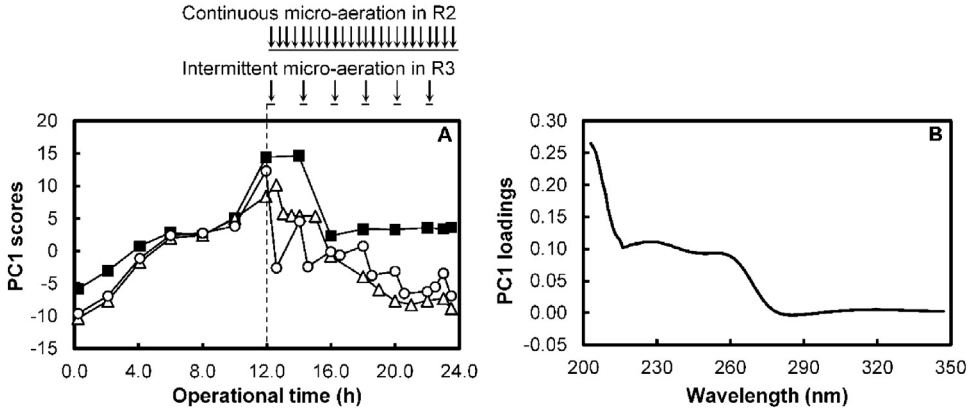


Fig. 5 – First principal component (PC1) scores (panel A) and loadings (panel B) of a PCA applied to spectral data (200–350 nm) of samples in a typical steady-state cycle (day 98) in R1 (—■—), R2 (—△—) and R3 (—○—). PC1 explained 73% of the spectral data variance, and was chosen as an indicator of aromatic amines formation and consumption.

targeted. The absorbance scans used as an input in PCA calculation and second principal component (PC2) results are shown in the Supplementary material. During the anaerobic phase of the cycle, the DB22 degradation (Fig. 4C) behavior was similar among the reactors and it was followed by the formation of aromatic amines (Fig. 5). However, after the beginning of micro-aeration in R2 and R3, a slight increase of the absorbance at 476 nm was detected (Fig. 4C). In R3, the absorbance at 476 nm increased and decreased following the pattern of air injection (Fig. 4C). The corresponding aromatic amines were then removed from R3 following an opposite pattern, as they underwent aerobic removal and anaerobic production (Fig. 5). On the other hand, in R2 a stable removal of amines was consequence of the continuous micro-aeration.

The absorbance increased at 476 nm detected in R2 and R3 from 12 h of cycle time was possibly due to the autoxidation of aromatic amines under oxygen exposure. Previous studies have described autoxidation as the process responsible for the color increase when reduced products from azo dye decolorization were exposed to oxygen (Barsing et al., 2011; Jonstrup et al., 2011). Autoxidation of aromatic amines can involve the removal of the hydrogen atoms from the amino group, releasing H_2O, so that an azo bond can be formed between the remaining nitrogen atoms of two different molecules. Considering that some aromatic amines exhibit more than one amino group, this reaction can go on, and ultimately lead to the formation of high-molecular-weight polymers with azo bonds (Olivares et al., 2016a, b). The autoxidized compounds are not susceptible to aerobic degradation and absorb light in the visible spectrum, thus increasing coloration (Khomenkov et al., 2008; Jonstrup et al., 2011). In general, autoxidation is a faster phenomenon than biological degradation of aromatic amines (Tan et al., 1999). Aromatic amines with hydroxyl groups, such as 3,6-diamino-4-hydroxynaphtalene-2-sulfonic acid (Fig. 3C) and 2,6-diamino-4-hydroxynaphtalene-1-sulfonic acid (Fig. 3D), are known to be rapidly autoxidized in the presence of air (Razo-Flores et al., 1997; Kudlich et al., 1999; Barsing et al., 2011). Besides, alkaline pH conditions also favor the autoxidation process (Josephy et al., 1983; Olivares et al., 2016b), as observed in R2 and R3 effluents (pH values of 8.1 ± 0.3 and 8.1 ± 0.2, respectively) compared to R1 effluent (pH values of 7.3 ± 0.2). This pH raise in the micro-aerated reactors was possibly due to the alkalinity production during aerobic biodegradation of organic matter.

It is necessary to highlight that in R3 the autoxidation products apparently were transiently converted to less colored intermediates during the anaerobic intervals (Figs. 4C and 5). Additionally, the autoxidation products were not able to cause inhibition in *V. fischeri* cells, since Microtox® did not detect any acute ecotoxicity in R2 and R3 effluents. This should be related to their high molecular weight, as previously discussed for DB22.

A peculiar decrease in aromatic amines concentration in R1 was observed between 14 and 16 h of the cycle time (Fig. 5). Contamination by oxygen was ruled out since redox potential did not significantly change during that specific interval (Fig. 4A). The possibility of anaerobic formation of azo dimers (Olivares et al., 2016a) was not followed by a coloration increase at 476 nm (Fig. 4C) or at any other wavelength of the visible spectrum (data not shown). Nevertheless, an unlikely anaerobic degradation of aromatic amines cannot be dismissed (Pereira et al., 2011; Sun et al., 2015), even though it has not been observed outside this interval.

In R2 and R3, the lowest absorbances at 476 nm were observed at the end of the anaerobic phase (12 h – Fig. 4C). This indicates that spectrophotometrically monitoring the absorbance of the final effluent underestimated the removal efficiency of DB22 by the micro-aerated reactors. That was due to the interference caused by autoxidation products, which can absorb light at the same characteristic wavelength.

Furthermore, it became evident that reductive decolorization was not an irreversible phenomenon, as also observed by Kudlich et al. (1999). Jonstrup et al. (2011) reported severe inefficiency in color removal due to autoxidation, which was not reproduced here. Absorbance increase by autoxidation in our study was not significant enough to make the combination of anaerobic and micro-aerated processes less efficient for color removal than the anaerobic process alone.

3.5.4. *Kinetic analysis*

The first order kinetic model fitted to DB22 decolorization (Table 1) indicated that the dye concentration was limiting the process, differently from what is reported for pure cultures in which zero order kinetic model reflects the dependence on the reducing equivalents production (van der Zee et al., 2001). In mixed anaerobic sludges, as herein, the production of reducing equivalents is faster and the limiting factor becomes the transference of electrons to the azo bonds.

At the anaerobic phase, R2 and R3 showed DB22 decolorization rates 73 and 123% higher than R1, respectively (from 0 to

Table 1 – Rate constants for DB22 and COD removal, and for estimate CH_4 production. Positive and negative values refer to production and removal, respectively. The coefficient of determination (R^2) is showed to assess how well the calculated rates fit the experimental data.

Reactor	DB22		COD (ethanol + VFA)	
Phase	k_1 (h^{-1}) First order	R^2	k_0 ($mg\,L^{-1}\,h^{-1}$) Order zero	R^2
R1				
Anaerobic phase (0–24 h)	−0.060	0.96	−49	0.86
R2				
Anaerobic phase (0–12 h)	−0.104	0.99	−32	0.99
Continuous micro-aerated phase (12–24 h)	N/A	N/A	−54	0.92
R3				
Anaerobic phase (0–12 h)	−0.134	0.97	−31	0.92
Intermittent micro-aerated phase (12–24 h)	N/A	N/A	−44	0.92

12 h for comparison). Nevertheless, COD was removed faster in R1 than in the other reactors during the anaerobic phase (0–12 h). R2 and R3 reached a higher rate of COD removal than R1 only during the micro-aerated stage. During the anaerobic intervals of the micro-aerated stage in R3, there was no effective degradation of COD, which decelerated the process (Table 1). Still, the micro-aeration in R2 was 4 times longer than in R3, but the consumption rate of COD was only 23% higher.

Dos Santos et al. (2006) pointed out that the reductive decolorization of azo dyes may compete with methanogenesis for the reducing equivalents produced by fermentative bacteria. Methanogenic inhibition by cyclic oxygen exposure could have favored the electrons transport to azo dye reduction, increasing the decolorization rates in R2 and R3. Furthermore, oxygen in those micro-aerated reactors was apparently more damaging to the methanogenesis than the accumulation of highly toxic aromatic amines in the anaerobic reactor was.

4. Conclusions

Both micro-aeration strategies tested improved the removal of aromatic amines and ecotoxicity, by alternating environments, compared to the single-stage anaerobic treatment. The continuous micro-aeration compared to the intermittent process resulted in a consumption of oxygen four times higher, but the overall performance was equivalent for both. Micro-aeration also apparently changed the balance between

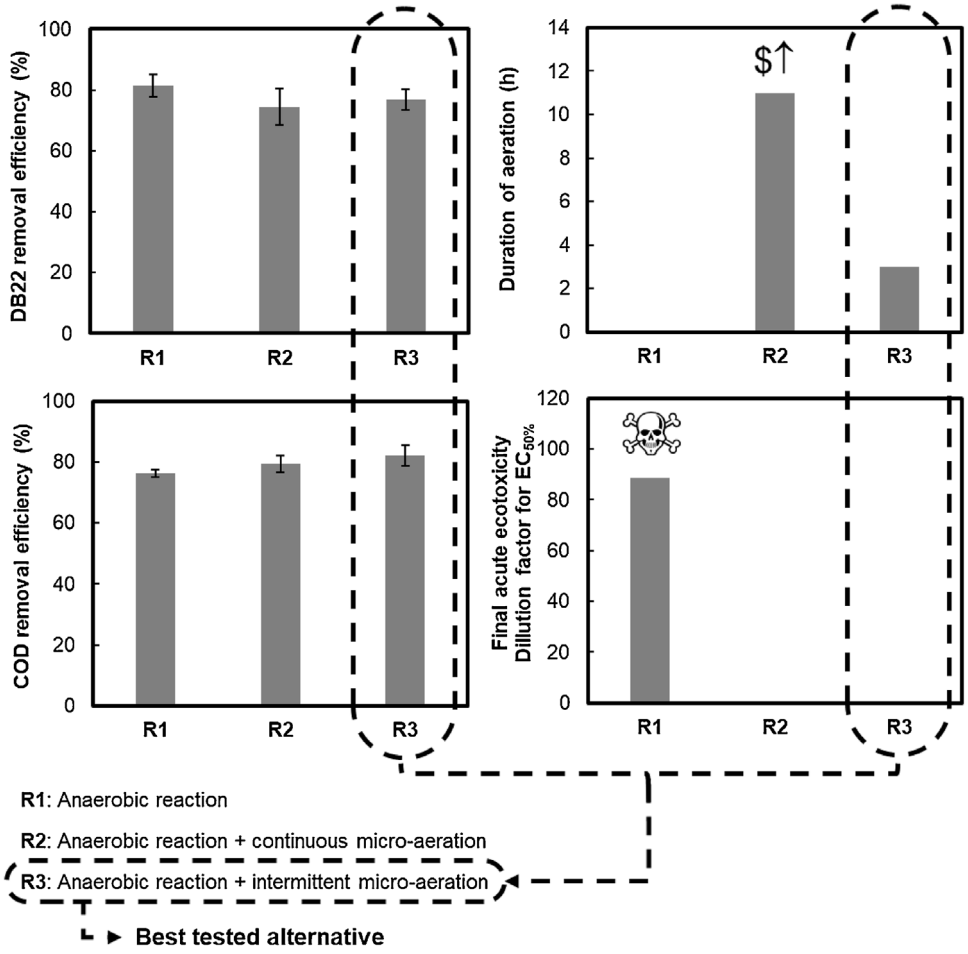


Fig. 6 – Summary showing the best tested alternative considering the main variables measured.

methanogenesis and reductive decolorization, favoring the second process. Methanogens seemed to be inhibited more by oxygen than by the accumulation of toxic aromatic amines. Overall, intermittent micro-aeration coupled to anaerobic digestion stood out as the better alternative (Fig. 6), showing COD and dye removal efficiencies higher than 76% and similar to those of the other two tested configurations; much lower oxygen consumption than that of continuous micro-aeration process; and a remarkable complete removal of acute ecotoxicity.

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References

Amaral, F.M., Kato, M.T., Florêncio, L., Gavazza, S., 2014. Color, organic matter and sulfate removal from textile effluents by anaerobic and aerobic processes. Bioresour. Technol. 163, 364–369, http://dx.doi.org/10.1016/j.biortech.2014.04.026.

Amaral, F.M., Florêncio, F., Kato, M.T., Santa-Cruz, P.A., Gavazza, S., 2017. Hydraulic retention time influence on azo dye and sulfate removal during the sequential anaerobic-aerobic treatment of real textile wastewater. Water Sci. Technol. 76, 3319–3327, http://dx.doi.org/10.2166/wst.2017.378.

Amorim, S.M., Kato, M.T., Florêncio, L., Gavazza, S., 2013. Influence of redox mediators and electron donors on the anaerobic removal of color and chemical oxygen demand from textile effluent. Clean Soil Air Water 41, 928–933, http://dx.doi.org/10.1002/clen.201200070.

APHA, 2005. Standard Methods for the Examination of Water & Wastewater. American Public Health Association, American Water Works Association, Water Environment Federation, Washington, D.C.

Arora, P.K., 2015. Bacterial degradation of monocyclic aromatic amine. Front. Microbiol. 6, 1–14, http://dx.doi.org/10.3389/fmicb.2015.00820.

Baêta, B.E.L., Lima, D.R.S., Silva, S.Q., Aquino, S.F., 2015. Evaluation of soluble microbial products and aromatic amines accumulation during a combined anaerobic/aerobic treatment of a model azo dye. Chem. Eng. J. 259, 936–944, http://dx.doi.org/10.1016/j.cej.2014.08.050.

Balapure, K., Jain, K., Bhatt, N., Madamwar, D., 2016. Exploring bioremediation strategies to enhance the mineralization of textile industrial wastewater through sequential anaerobic-microaerophilic process. Int. Biodeter. Biodegr. 106, 97–105, http://dx.doi.org/10.1016/j.ibiod.2015.10.008.

Barsing, P., Tiwari, A., Joshi, T., Garg, S., 2011. Application of a novel bacterial consortium for mineralization of sulphonated aromatic amines. Bioresour. Technol. 102, 765–771, http://dx.doi.org/10.1016/j.biortech.2010.08.098.

Beebe, K.R., Pell, R.J., Seasholtz, M.B., 1998. Chemometrics: A Practical Guide. Wiley-Interscience, New York, NY.

Bro, R., Smilde, A.K., 2014. Principal component analysis. Anal. Methods 6, 2812–2831, http://dx.doi.org/10.1039/c3ay41907j.

da Silva, M.E.R., Firmino, P.I.M., de Sousa, M.R., dos Santos, A.B., 2012. Sequential anaerobic/aerobic treatment of dye-containing wastewaters: colour and COD removals, and ecotoxicity tests. Appl. Biochem. Biotechnol. 166, 1057–1069, http://dx.doi.org/10.1007/s12010-011-9493-7.

dos Santos, A.B., de Madrid, M.P., de Bok, F.A.M., Stams, A.J.M., van Lier, J.B., Cervantes, F.J., 2006. The contribution of fermentative bacteria and methanogenic archaea to azo dye reduction by a thermophilic anaerobic consortium. Enzyme Microb. Technol. 39, 38–46, http://dx.doi.org/10.1016/j.enzmictec.2005.09.003.

dos Santos, A.B., Cervantes, F.J., van Lier, J.B., 2007. Review paper on current technologies for decolourisation of textile wastewaters: perspectives for anaerobic biotechnology. Bioresour. Technol. 98, 2369–2385, http://dx.doi.org/10.1016/j.biortech.2006.11.013.

Farabegoli, G., Chiavola, A., Rolle, E., Naso, M., 2010. Decolorization of Reactive Red 195 by a mixed culture in an alternating anaerobic-aerobic Sequencing Batch Reactor. Biochem. Eng. J. 52, 220–226, http://dx.doi.org/10.1016/j.bej.2010.08.014.

Ferraz, E.R.A., Grando, M.D., Oliveira, D.P., 2011. The azo dye Disperse Orange 1 induces DNA damage and cytotoxic effects but does not cause ecotoxic effects in *Daphnia similis* and *Vibrio fischeri*. J. Hazard. Mater. 192, 628–633, http://dx.doi.org/10.1016/j.jhazmat.2011.05.063.

Field, J.A., Stams, A.J.M., Kato, M., Schraa, G., 1995. Enhanced biodegradation of aromatic pollutants in cocultures of anaerobic and aerobic bacterial consortia. Antonie Van Leeuwenhoek 67, 47–77, http://dx.doi.org/10.1007/BF00872195.

Florêncio, L., Jenicek, P., Field, J.A., Lettinga, G., 1993. Effect of cobalt on the anaerobic degradation of methanol. J. Ferment. Bioeng. 75, 368–374, http://dx.doi.org/10.1016/0922-338X(93)90136-V.

Franciscon, E., Zille, A., Fantinatti-Garboggini, F., Silva, I.S., Cavaco-Paulo, A., Durrant, L.R., 2009a. Microaerophilic-aerobic sequential decolourization/biodegradation of textile azo dyes by a facultative *Klebsiella* sp. strain VN-31. Process Biochem. 44, 446–452, http://dx.doi.org/10.1016/j.procbio.2008.12.009.

Franciscon, E., Zille, A., Guimaro, F.D., de Menezes, C.R., Durrant, L.R., Cavaco-Paulo, A., 2009b. Biodegradation of textile azo dyes by a facultative *Staphylococcus arlettae* strain VN-11 using a sequential microaerophilic/aerobic process. Int. Biodeterior. Biodegrad. 63, 280–288, http://dx.doi.org/10.1016/j.ibiod.2008.10.003.

Frijters, C.T.M.J., Vos, R.H., Scheffer, G., Mulder, R., 2006. Mulder, decolorizing and detoxifying textile wastewater, containing both soluble and insoluble dyes, in a full scale combined anaerobic/aerobic system. Water Res. 40, 1249–1257, http://dx.doi.org/10.1016/j.watres.2006.01.013.

Fuchs, G., Boll, M., Heider, J., 2011. Microbial degradation of aromatic compounds — from one strategy to four. Nat. Rev. Microbiol. 9, 803–816, http://dx.doi.org/10.1038/nrmicro2652.

Gavazza, S., Guzman, J.J.L., Angenent, L.T., 2015. Electrolysis within anaerobic bioreactors stimulates breakdown of toxic products from azo dye treatment. Biodegradation 26, 151–160, http://dx.doi.org/10.1007/s10532-015-9723-8.

Gottlieb, A., Shaw, C., Smith, A., Wheatley, A., Forsythe, S., 2003. The toxicity of textile reactive azo dyes after hydrolysis and decolourisation. J. Biotechnol. 101, 49–56, http://dx.doi.org/10.1016/S0168-1656(02)00302-4.

Harper, S.R., Pohland, F.G., 1986. Recent developments in hydrogen management during anaerobic biological wastewater treatment. Biotechnol. Bioeng. 18, 585–602, http://dx.doi.org/10.1007/Bfb0000691.

Jonstrup, M., Kumar, N., Murto, M., Mattiasson, B., 2011. Sequential anaerobic-aerobic treatment of azo dyes: decolourisation and amine degradability. Desalination 280, 339–346, http://dx.doi.org/10.1016/j.desal.2011.07.022.

Josephy, P.D., Eling, T.E., Mason, R.P., 1983. Oxidation of p-aminophenol catalyzed by horseradish peroxidase and prostaglandin synthase. Mol. Pharmacol. 23, 461–466.

Kato, M.T., Field, J.A., Lettinga, G., 1997. Anaerobe tolerance to oxygen and the potentials of anaerobic and aerobic cocultures for wastewater treatment. Braz. J. Chem. Eng. 14, 1–12, http://dx.doi.org/10.1590/S0104-66321997000400015.

Khan, R., Bhawana, P., Fulekar, M.H., 2013. Microbial decolorization and degradation of synthetic dyes: a review. Rev. Environ. Sci. Biotechnol. 12, 75–97, http://dx.doi.org/10.1007/s11157-012-9287-6.

Khomenkov, V.G., Shevelev, A.B., Zhukov, V.G., Zagustina, A.M., Bezborodov, A.M., Popov, V.O., 2008. Organization of metabolic pathways and molecular-genetic mechanisms of xenobiotic degradation in microorganisms: a review. Appl. Biochem. Microbiol. 44, 117–135, http://dx.doi.org/10.1134/S0003683808020014.

Kudlich, M., Hetheridge, M.J., Knackmuss, H.J., Stolz, A., 1999. Autoxidation reactions of different aromatic o-aminohydroxynaphthalenes that are formed during the anaerobic reduction of sulfonated azo dyes. Environ. Sci. Technol. 33, 896–901, http://dx.doi.org/10.1021/es9808346.

Kulla, H.G., Klausener, F., Meyer, U., Lüdeke, B., Leisinger, T., 1983. Interference of aromatic sulfo groups in the microbial degradation of the azo dyes Orange I and Orange II. Arch. Microbiol. 135, 1–7, http://dx.doi.org/10.1007/BF00419473.

Kumar, A.N., Reddy, C.N., Mohan, S.V., 2015. Biomineralization of azo dye bearing wastewater in periodic discontinuous batch reactor: effect of microaerophilic conditions on treatment efficiency. Bioresour. Technol. 188, 56–64, http://dx.doi.org/10.1016/j.biortech.2015.01.098.

Lourenço, N.D., Novais, J.M., Pinheiro, H.M., 2001. Effect of some operational parameters on textile dye biodegradation in a sequential batch reactor. J. Biotechnol. 89, 163–174, http://dx.doi.org/10.1016/S0168-1656(01)00313-3.

Muyzer, G., Stams, A.J.M., 2008. The ecology and biotechnology of sulphate-reducing bacteria. Nat. Rev. Microbiol. 6, 441–454, http://dx.doi.org/10.1038/nrmicro1892.

Olivares, C.I., Abrell, L., Khatiwada, R., Chorover, J., Sierra-Alvarez, R., Field, J.A., 2016a. (Bio)transformation of 2,4-dinitroanisole (DNAN) in soils. J. Hazard. Mater. 304, 214–221, http://dx.doi.org/10.1016/j.jhazmat.2015.10.059.

Olivares, C.I., Wang, J., Silva Luna, C.D., Field, J.A., Abrell, L., Sierra-Alvarez, R., 2016b. Continuous treatment of the insensitive munitions compound N-methyl-p-nitro aniline (MNA) in an upflow anaerobic sludge blanket (UASB) bioreactor. Chemosphere 144, 1116–1122, http://dx.doi.org/10.1016/j.chemosphere.2015.09.092.

Ong, S.A., Toorisaka, E., Hirata, M., Hano, T., 2005. Decolorization of azo dye (Orange II) in a sequential UASB-SBR system. Sep. Purif. Technol. 42, 297–302, http://dx.doi.org/10.1016/j.seppur.2004.09.004.

Pagga, U., Brown, D., 1986. The degradation of dyestuffs: part II Behaviour of dyestuffs in aerobic biodegradation tests. Chemosphere 15, 479–491, http://dx.doi.org/10.1016/0045-6535(86)90542-4.

Pereira, R., Pereira, L., van der Zee, F.P., Alves, M.M., 2011. Fate of aniline and sulfanilic acid in UASB bioreactors under denitrifying conditions. Water Res. 45, 191–200, http://dx.doi.org/10.1016/j.watres.2010.08.027.

Pinheiro, M.H., Touraud, E., Thomas, O., 2004. Aromatic amines from azo dye reduction: status review with emphasis on direct UV spectrophotometric detection in textile industry wastewaters. Dyes Pigm. 61, 121–139, http://dx.doi.org/10.1016/j.dyepig.2003.10.009.

Rawat, D., Mishra, V., Sharma, R.S., 2016. Detoxification of azo dyes in the context of environmental processes. Chemosphere 155, 591–605, http://dx.doi.org/10.1016/j.chemosphere.2016.04.068.

Razo-Flores, E., Luijten, M., Donlon, B.A., Lettinga, G., Field, J.A., 1997. Complete biodegradation of the azo dye azodisalicylate under anaerobic conditions. Environ. Sci. Technol. 31, 2098–2103, http://dx.doi.org/10.1021/es960933o.

Shaul, G.M., Holdsworth, T.J., Dempsey, C.R., Dostal, K.A., 1991. Fate of water soluble azo dyes in the activated sludge process. Chemosphere 22, 107–119, http://dx.doi.org/10.1016/0045-6535(91)90269-J.

Sponza, D.T., Işik, M., 2005. Toxicity and intermediates of C.I. Direct Red 28 dye through sequential anaerobic/aerobic treatment. Process Biochem. 40, 2735–2744, http://dx.doi.org/10.1016/j.procbio.2004.12.016.

Sun, W., Li, Y., McGuinness, L.R., Luo, S., Huang, W., Kerkhof, L.J., Mack, E.E., Häggblom, M.M., Fennell, D.E., 2015. Identification of anaerobic aniline-degrading Bacteria at a contaminated industrial site. Environ. Sci. Technol. 49, 11079–11088, http://dx.doi.org/10.1021/acs.est.5b02166.

Tan, N.C.G., Prenafeta-Boldú, F.X., Opsteeg, J.L., Lettinga, G., Field, J.A., 1999. Biodegradation of azo dyes in cocultures of anaerobic granular sludge with aerobic aromatic amine degrading enrichment cultures. Appl. Microbiol. Biotechnol. 51, 865–871, http://dx.doi.org/10.1007/s002530051475.

van der Zee, F.P., Villaverde, S., 2005. Combined anaerobic-aerobic treatment of azo dyes — a short review of bioreactor studies. Water Res. 39, 1425–1440, http://dx.doi.org/10.1016/j.watres.2005.03.007.

van der Zee, F.P., Lettinga, G., Field, J.A., 2001. Azo dye decolourisation by anaerobic granular sludge. Chemosphere 44, 1169–1176, http://dx.doi.org/10.1016/S0045-6535(00)00270-8.

Yaşar, S., Cirik, K., Çinar, Ö., 2012. The effect of cyclic anaerobic-aerobic conditions on biodegradation of azo dyes. Bioprocess Biosyst. Eng. 35, 449–457, http://dx.doi.org/10.1007/s00449-011-0584-1.